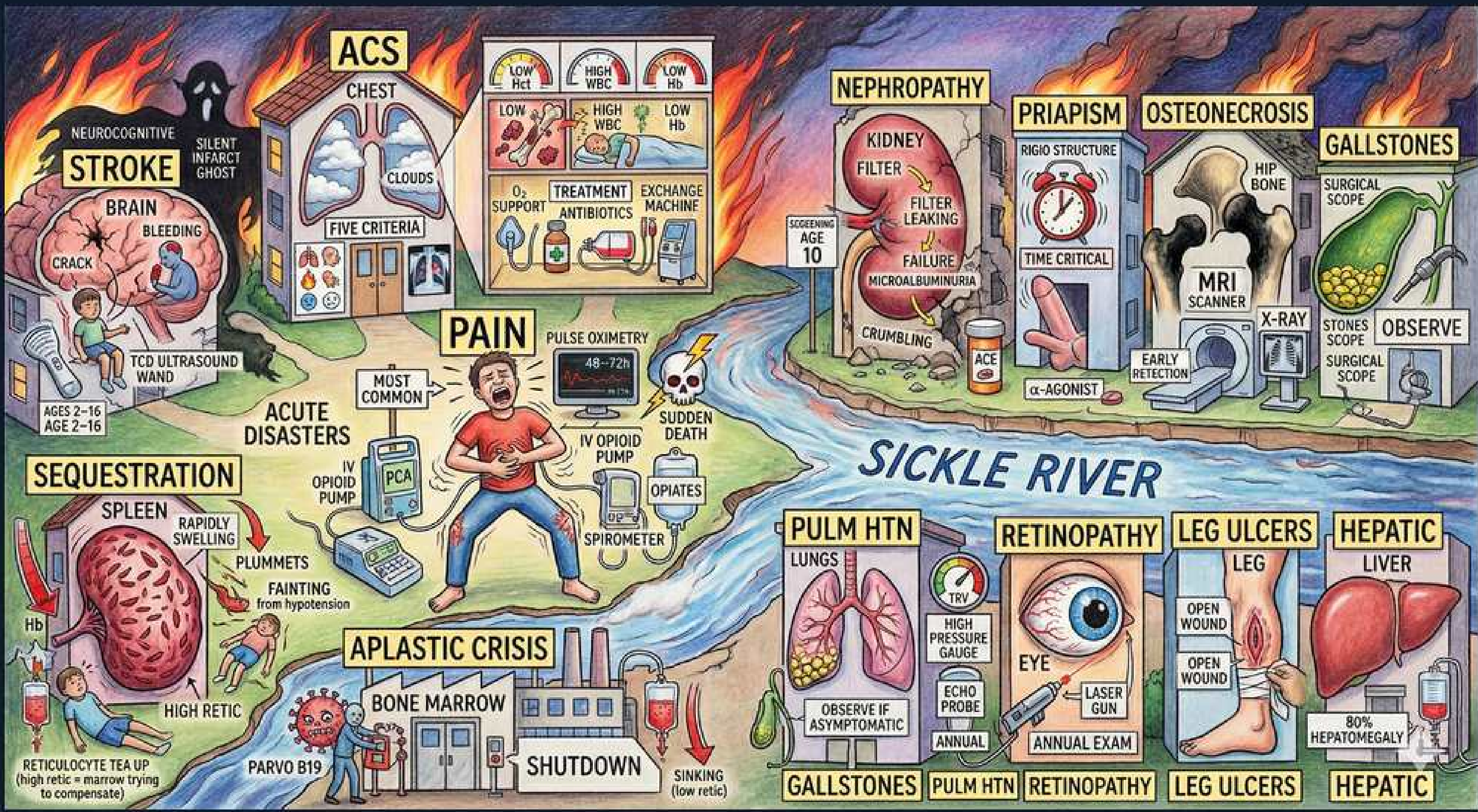
Hemoglobinopathy — The Sickle River: Catastrophes



1. Vaso-occlusive pain crisis

WHAT YOU SEE

Patient screaming, IV PCA opioid pump, 'MOST COMMON'

WHAT IT ENCODES

Vaso-occlusive crisis (VOC) is the MOST COMMON acute event and the #1 reason for ED visits/hospitalization, driven by sickled RBCs plus adherent leukocytes/platelets occluding the microvasculature → ischemic bone/marrow pain. Treat with prompt IV opioids (PCA morphine/hydromorphone), aggressive hydration, and oxygen for hypoxia; add NSAIDs and incentive spirometry to prevent ACS. Routine transfusion is NOT indicated for an uncomplicated pain crisis. Common triggers: cold, dehydration, infection, hypoxia, stress.

BOARD TRAP

A board stem describing a typical limb/back pain crisis tempts you to 'transfuse to relieve the crisis' — wrong; simple transfusion does not abort VOC pain. The right answer is analgesia + hydration + O2. Reserve transfusion for ACS, stroke, sequestration, or symptomatic severe anemia.

---

2. Stroke

WHAT YOU SEE

Brain with crack + bleeding; TCD ultrasound wand, ages 2-16

WHAT IT ENCODES

Stroke is a leading cause of disability in SCD; children get ischemic (large-vessel/moyamoya) strokes while adults more often have hemorrhagic strokes. Screen ALL HbSS children ages 2–16 yearly with transcranial Doppler (TCD); a time-averaged mean velocity ≥200 cm/s is high-risk. Primary prevention for high TCD = chronic transfusion to keep HbS <30%; hydroxyurea can substitute once stabilized (TWiTCH trial). Acute stroke → urgent EXCHANGE transfusion.

BOARD TRAP

Stem gives a high-velocity TCD in an 8-year-old and asks the next step; the distractor 'start hydroxyurea' or 'MRI and observe' is wrong for PRIMARY prevention — the evidence-based answer (STOP trial) is begin chronic transfusion to lower HbS. Hydroxyurea is for after transfusion or where transfusion isn't feasible.

---

3. Acute chest syndrome

WHAT YOU SEE

Chest with clouds/infiltrate; 'FIVE CRITERIA', exchange machine

WHAT IT ENCODES

Acute chest syndrome (ACS) = NEW pulmonary infiltrate plus fever and/or respiratory symptoms (cough, chest pain, hypoxia, tachypnea); it is the #1 cause of death in SCD. Causes: fat embolism from infarcted marrow, infection (Chlamydia/Mycoplasma/viral/pneumococcus), and in-situ pulmonary vaso-occlusion. Treat with O2, empiric antibiotics covering atypicals (ceftriaxone + azithromycin), bronchodilators, pain control + incentive spirometry, and TRANSFUSION — exchange transfusion for severe/multilobar disease.

BOARD TRAP

Classic trap: the crisis patient on opioids who then becomes hypoxic with a new infiltrate is mislabeled as 'opioid oversedation/pneumonia, just give antibiotics.' Discriminator: any new infiltrate + respiratory symptoms in SCD = ACS, which mandates urgent transfusion (exchange if severe), not antibiotics alone. Also do not call VOC the leading cause of death — ACS (and infection) is.

---

4. Splenic sequestration

WHAT YOU SEE

Spleen rapidly swelling, Hb plummets, fainting, ↑retic

WHAT IT ENCODES

Acute splenic sequestration = sudden pooling of blood in the spleen → rapidly enlarging tender spleen, a precipitous Hgb drop (often >2 g/dL), and hypovolemic shock; reticulocyte count is HIGH (marrow is responding). It strikes young HbSS children (before autosplenectomy) and HbSC/Sβ+ patients at any age. Treat with cautious transfusion/volume resuscitation; splenectomy after recurrent episodes.

BOARD TRAP

Trap: a toddler with a falling Hgb is labeled 'aplastic crisis.' Discriminator = the reticulocyte count: sequestration has HIGH retics and a palpable enlarging spleen, whereas parvovirus aplastic crisis has LOW retics. Over-transfusing is another pitfall — blood is autotransfused back as the spleen contracts, risking hyperviscosity.

---

5. Aplastic crisis — Parvo B19

WHAT YOU SEE

Bone marrow 'SHUTDOWN' factory, Parvo B19, retic tea cup empty

WHAT IT ENCODES

Aplastic crisis = Parvovirus B19 infects erythroid progenitors and halts erythropoiesis for ~7–10 days → sharp Hgb fall with an inappropriately LOW reticulocyte count (<1%). Because SCD red cells live only ~10–20 days, even brief marrow shutdown causes severe anemia. Self-limited; treat with transfusion until marrow recovers; IVIG in chronically infected/immunocompromised.

BOARD TRAP

Trap answer: 'high reticulocytes confirm aplastic crisis' — WRONG; the hallmark is a LOW retic count because the marrow is shut down. High retics with a big spleen = sequestration; low retics with normal-size spleen + parvovirus exposure (or a sick contact with 'slapped-cheek' rash) = aplastic crisis.

---

6. Sudden death

WHAT YOU SEE

Skull icon labeled 'SUDDEN DEATH'

WHAT IT ENCODES

SCD carries excess sudden-death risk from acute chest syndrome, massive splenic sequestration, overwhelming sepsis from encapsulated organisms (autosplenectomy), fat/pulmonary embolism, and arrhythmia from pulmonary hypertension/iron-overload cardiomyopathy. Median survival, though improved, remains reduced. Monitor pulse oximetry and pulmonary function; aggressive treatment of ACS, fever, and PH lowers mortality.

BOARD TRAP

Easy to attribute a sudden SCD death to 'narcotic overdose' from crisis analgesia; the more likely board answer is ACS or fulminant sepsis. Any febrile SCD patient is a medical emergency (functional asplenia) — failure to give empiric antibiotics promptly is the tested error.

---

7. Nephropathy

WHAT YOU SEE

Kidney filter failing, microalbuminuria, ACEi

WHAT IT ENCODES

Sickle nephropathy begins with the hypoxic, acidotic, hypertonic renal medulla: sickling there causes papillary necrosis (painless hematuria) and an inability to concentrate urine (hyposthenuria, nocturia). Progresses through hyperfiltration → microalbuminuria → proteinuria → CKD/ESRD. Treat proteinuria with ACE inhibitors/ARBs; hydroxyurea may help. Renal medullary CARCINOMA is associated with sickle TRAIT.

BOARD TRAP

Trap: gross painless hematuria in a sickle-TRAIT athlete is called 'glomerulonephritis/IgA.' Discriminator: think renal papillary necrosis (or rule out renal medullary carcinoma, which is the classic malignancy of sickle trait). Hyposthenuria — not nephrotic-range disease — is the earliest, most universal renal finding.

---

8. Priapism

WHAT YOU SEE

Rigid structure, 'TIME CRITICAL', α-agonist

WHAT IT ENCODES

Priapism is a vaso-occlusive urologic EMERGENCY (low-flow/ischemic type). Treatment is time-critical: an episode >4 hours requires corporal aspiration of blood and intracavernosal injection of an α-adrenergic agonist (phenylephrine); add hydration, analgesia, and consider exchange transfusion for refractory cases. Recurrent or prolonged episodes cause fibrosis and permanent erectile dysfunction.

BOARD TRAP

Trap: treating ischemic priapism like a benign issue or jumping to surgery first; the discriminator is that low-flow priapism is ischemic and time-critical — aspiration + intracavernosal phenylephrine is first-line, and delay (>4 h) risks impotence. (High-flow/non-ischemic priapism, by contrast, is not an emergency.)

---

9. Osteonecrosis (AVN)

WHAT YOU SEE

Hip bone, MRI scanner, X-ray

WHAT IT ENCODES

Avascular (aseptic) necrosis follows bone infarction, classically of the femoral head (also humeral head) → chronic hip pain and limp. MRI is the most sensitive test for EARLY AVN before plain films change; X-ray later shows sclerosis, subchondral collapse, 'crescent sign.' Manage with analgesia, physical therapy, core decompression early, joint replacement when collapsed.

BOARD TRAP

Trap: hip pain in SCD with a normal X-ray is read as 'no pathology' or assumed to be septic arthritis/osteomyelitis. Discriminator: a normal radiograph does not exclude early AVN — order MRI. Distinguish AVN (ischemic bone infarct, sterile) from osteomyelitis (fever, ↑ESR/CRP, Salmonella the classic organism).

---

10. Pulmonary HTN

WHAT YOU SEE

Lungs with high-pressure gauge, echo probe annual

WHAT IT ENCODES

Chronic intravascular hemolysis releases free Hgb and arginase that scavenge/deplete nitric oxide → pulmonary vascular remodeling and pulmonary hypertension, part of the hemolysis-endothelial phenotype (with leg ulcers, priapism, stroke). Screen with echocardiography; a tricuspid regurgitant (TR) jet velocity ≥2.5 m/s predicts increased mortality and warrants right-heart catheterization. PH is a strong independent predictor of death in SCD.

BOARD TRAP

Trap: an elevated TR jet on echo is dismissed as 'incidental.' Discriminator: TR jet ≥2.5 m/s is a powerful independent mortality predictor and should trigger right-heart cath to confirm PH, not reassurance.

---

11. Retinopathy

WHAT YOU SEE

Eye, laser icon, annual exam

WHAT IT ENCODES

Proliferative sickle retinopathy arises from peripheral retinal ischemia → neovascularization ('sea-fan' vessels) that can bleed (vitreous hemorrhage) or cause tractional retinal detachment and vision loss. It is paradoxically MORE common and severe in HbSC and Sβ+ disease than in HbSS. Screen with annual dilated fundoscopy starting ~age 10; treat with laser photocoagulation/anti-VEGF.

BOARD TRAP

Trap: assuming the sickest genotype (HbSS) has the worst eye disease. Discriminator: proliferative retinopathy is classically WORSE in HbSC disease — a frequently tested 'milder overall but eye-aggressive' pattern. Skipping annual dilated exams in HbSC is the management error.

---

12. Leg ulcers

WHAT YOU SEE

Leg with open wounds

WHAT IT ENCODES

Chronic, painful malleolar (medial/lateral ankle) leg ulcers result from the hemolysis/NO-depletion phenotype plus poor microvascular perfusion and venous insufficiency. They are indolent, recur, and heal slowly. Management: compression, wound care, treat infection, optimize hydroxyurea; transfusion for refractory ulcers. Associated with low Hgb and high hemolytic markers (LDH).

BOARD TRAP

Trap: a chronic ankle ulcer in a young SCD patient is worked up only as 'diabetic' or 'pure venous stasis' ulcer. Discriminator: in SCD the classic site is the malleolar region and the driver is the hemolytic phenotype — these cluster with pulmonary HTN, priapism, and leg ulcers, not with the viscosity/VOC phenotype.

---

13. Hepatic / gallstones

WHAT YOU SEE

Liver, 80% hepatomegaly, gallstones

WHAT IT ENCODES

Chronic extravascular hemolysis raises unconjugated bilirubin → pigment (calcium bilirubinate) GALLSTONES, often by adolescence (prevalence ~70–80% with age); cholecystectomy for symptomatic stones. The liver can also suffer acute hepatic sequestration/crisis and intrahepatic cholestasis, and transfusion-dependent patients develop iron overload (treat with chelation: deferasirox/deferoxamine).

BOARD TRAP

Trap: RUQ pain + jaundice in SCD is attributed to 'hemolysis alone, no imaging needed.' Discriminator: pigment gallstones are extremely common — get a RUQ ultrasound; rising direct bilirubin/markedly elevated transaminases suggest hepatic crisis or choledocholithiasis. In chronically transfused patients, ferritin-driven iron overload (not new hemolysis) is the cause of organ damage.